

Comment on AI Action Plan RFI – Advancing AI Leadership through Cloud Infrastructure, Regulatory Modernization, and Immigration Reform

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Submitted by: Summit Singh Thakur

AI & Cloud Infrastructure Expert | Startup & Innovation Strategist

Executive Summary

The United States finds itself at a critical inflection point in the global artificial intelligence (AI) ecosystem. The ability to maintain a leadership position in AI-driven innovation, economic competitiveness, and national security hinges upon decisive policy actions that address structural inefficiencies in infrastructure, regulatory frameworks, and talent acquisition. Current trends indicate a marked decline in AI PhD retention rates in the U.S., with approximately 70% of international graduates seeking employment opportunities abroad due to restrictive immigration policies and more lucrative incentives in competing markets such as China, the European Union, and Canada. The Center for Security and Emerging Technology warns that this intellectual exodus poses significant risks to the nation's AI preeminence, potentially undermining long-term strategic objectives. To counteract these trends and reinforce the U.S. position as the global nucleus of AI advancement, this paper delineates a comprehensive strategy to:

- **Enhance AI-driven cloud infrastructure** by optimizing computational efficiency, reducing latency, and expanding high-performance computing capabilities.
- **Dismantle regulatory bottlenecks** that stifle AI innovation while implementing ethical oversight mechanisms to prevent misuse and bias.
- **Streamline immigration pathways** to attract and retain elite AI researchers, engineers, and entrepreneurs who drive transformative advancements.
- **Establish robust AI security frameworks** that address algorithmic vulnerabilities, reinforce cybersecurity measures, and ensure data integrity.
- **Foster cross-disciplinary collaboration** between academia, industry, and government to accelerate groundbreaking AI research and commercialization.

- **Expand workforce development initiatives** to cultivate a robust domestic pipeline of AI talent and reduce dependence on foreign expertise.
- **Implement ethical AI governance structures** that ensure responsible AI deployment while mitigating algorithmic bias and unintended consequences.
- **Align federal and state AI initiatives** to create a unified national strategy that fosters sustained AI growth and technological sovereignty.

Through a multifaceted and proactive policy approach, the AI Action Plan can solidify the United States' leadership in AI while bolstering economic prosperity, reinforcing national security, and safeguarding technological sovereignty.

Optimizing AI Infrastructure for Scalable Innovation

The rapid evolution of AI applications necessitates a robust technological backbone characterized by high-performance computing (HPC), expansive data storage capacity, and low-latency network infrastructure. To maintain computational supremacy, the U.S. must implement strategic interventions in the following domains:

1.1 Strategic Investment in AI-Optimized Cloud Computing

- Increase federal and private-sector investment in AI-accelerated cloud architectures to facilitate large-scale AI model deployment with heightened efficiency.
- Introduce tax incentives and targeted research grants to support AI-driven data centers that integrate sustainable energy solutions, following successful implementations in Sweden and Singapore.
- Enhance access to federal AI supercomputing resources for startups, research institutions, and academic organizations to democratize high-performance computing capabilities.
- Develop AI-centric cloud ecosystems tailored to the unique computational demands of enterprises leveraging advanced machine learning techniques.
- Expand the integration of AI models with edge computing frameworks to enable real-time inference for mission-critical applications.

- Cultivate strategic alliances between cloud service providers and AI research institutions to optimize computational methodologies and improve cost efficiency.

1.2 AI-Specific Semiconductor and GPU Innovation

- Intensify federal investments in domestic semiconductor fabrication initiatives, with an emphasis on AI-optimized processors, under expanded provisions of the CHIPS Act.
 - Reduce reliance on foreign semiconductor supply chains by incentivizing domestic AI chip manufacturing and fostering industry-led innovation in AI accelerators.
 - Strengthen public-private partnerships aimed at pioneering next-generation AI chipsets designed for superior computational efficiency and sustainability.
 - Drive research and development (R&D) in neuromorphic computing and quantum AI to unlock disruptive advancements in AI processing paradigms.
 - Provide fiscal incentives for AI hardware startups to enhance domestic semiconductor manufacturing capacity and mitigate geopolitical risks.
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Removing Structural Barriers to AI Startup Growth

AI startups serve as critical engines of technological progress but frequently encounter prohibitive regulatory constraints, inconsistent funding mechanisms, and bureaucratic inefficiencies. The AI Action Plan must prioritize:

2.1 Catalyzing Public-Private AI Innovation Partnerships

- Establish federally funded AI R&D initiatives modeled after DARPA's innovation ecosystem, fostering high-impact collaborations between academia and industry leaders.
- Expedite federal grant allocations and introduce targeted tax incentives for AI-driven startups in high-priority sectors such as national security, precision medicine, and financial technology.
- Launch a National AI Innovation Fund to provide targeted capital support to early-stage AI enterprises demonstrating transformative potential.

- Develop a centralized AI startup registry to streamline access to funding, compliance support, and mentorship programs.
- Strengthen the synergy between AI startups and government technology incubators to expedite the commercialization of pioneering AI solutions.

2.2 Streamlining AI Compliance to Expedite Deployment

- Harmonize AI regulatory frameworks to replace fragmented state-level policies with a cohesive national standard, ensuring regulatory clarity and consistency.
 - Reduce bureaucratic inertia in AI approval processes across critical industries, particularly in finance, healthcare, and cybersecurity.
 - Facilitate ongoing collaboration between regulatory agencies (e.g., FTC, NIST, Department of Commerce) and AI experts to establish adaptive policies that balance innovation with ethical considerations.
 - Implement AI regulatory sandboxes that enable real-world testing of AI models under controlled conditions before full-scale deployment.
 - Create a real-time AI regulation advisory board to provide dynamic policy adjustments in response to rapid technological advancements.
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Immigration Reforms to Retain AI Talent

To sustain its AI leadership, the U.S. must enact forward-looking immigration policies that enhance talent retention and facilitate the recruitment of world-class AI professionals.

3.1 Expanding EB-1A Eligibility for AI Experts

- Redefine AI startup founders and leading AI researchers as individuals of extraordinary ability to streamline their eligibility for EB-1A visas.
- Introduce an AI-specific visa category akin to Canada's Global Talent Stream, aimed at attracting highly skilled AI professionals.
- Expand the National Interest Waiver (NIW) framework to recognize AI entrepreneurs contributing to national strategic priorities.

- Foster AI talent exchange programs with allied nations to enhance global AI research collaboration and knowledge transfer.

3.2 Expedited Green Card Pathways for AI Innovators

- Establish priority processing mechanisms for AI entrepreneurs backed by venture capital, government grants, or high-impact research initiatives.
 - Permit concurrent filing of I-140 petitions and adjustment-of-status applications for AI professionals to minimize prolonged processing delays.
 - Broaden O-1 visa criteria to encompass a wider spectrum of AI professionals engaged in groundbreaking research and development.
 - Develop a direct-to-green-card program for AI PhD graduates actively engaged in critical AI sectors.
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Strengthening AI Security and Data Governance

The increasing sophistication of AI necessitates the implementation of rigorous security protocols to mitigate cyber threats, algorithmic vulnerabilities, and data privacy concerns.

4.1 National AI Security Standards

- Mandate zero-trust security models for AI infrastructure deployments in cloud-based environments.
- Enforce federal cybersecurity protocols to safeguard AI models against adversarial manipulation, deepfake proliferation, and AI-powered cyber threats.
- Develop a national AI risk assessment framework to systematically evaluate vulnerabilities across AI-dependent industries.
- Invest in AI-driven cybersecurity solutions capable of preemptively detecting and neutralizing emerging threats.

4.2 Comprehensive AI Data Privacy and Cybersecurity Frameworks

- Implement stringent AI data governance policies to prevent unauthorized data access, mitigate biases in training datasets, and ensure AI model integrity.

- Require cloud providers to publish transparency reports detailing AI-related security incidents, ethical risks, and bias mitigation efforts.

Conclusion

By strategically investing in AI infrastructure, reforming regulatory constraints, modernizing immigration pathways, and strengthening AI security, the U.S. can secure its leadership position in the global AI race while fostering long-term economic growth and national security.

Submission Details:

Summit Singh Thakur

Co-Founder, CTO at TruckBux, Inc.

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